

Homework 1

Alejandro De La Torre
MATH 431 — Introduction to Probability
LEC 002
Instructor: Sarah Tammen

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Problem 1.4 (*Sample space and probability*)

State the problem here (paste from PDF).

Solution.

We model the experiment using a probability space

$$(\Omega, \mathcal{F}, P).$$

Scratch / Setup. The sample space is

$$\Omega = \{H, T\}^3,$$

the set of all length-3 sequences of coin flips. The σ -field is the power set $\mathcal{F} = 2^\Omega$.

Assuming the coin is fair, we assign equal probability to each outcome:

$$P(\{\omega\}) = 2^{-3} \quad \text{for all } \omega \in \Omega.$$

Let

$$A = \{\omega \in \Omega : \text{exactly two heads occur}\}.$$

There are $\binom{3}{2} = 3$ such outcomes, so $|A| = 3$ and

$$|\Omega| = 2^3 = 8.$$

Since all outcomes are equally likely,

$$P(A) = \frac{|A|}{|\Omega|} = \frac{3}{8}.$$

Final: $P(A) = \frac{3}{8}$